PM_Inv - User Manual

PM_Inv: Predominant-Mode Surface-Wave Inversion Using Differential Evolution
Version 1.0 (2026)

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1. Introduction

PM_Inv is a MATLAB-based software package for 1-D surface-wave inversion using the **Predominant Mode (PM) inversion framework**. The software estimates layered shear-wave velocity (Vs) profiles from experimentally measured surface-wave phase velocity dispersion curves by coupling a **Hybrid Thin-Layer Method (HTLM)** forward solver with an adaptive **Differential Evolution (DE)** optimization algorithm.

Unlike conventional multimodal inversion approaches that require manual mode identification and mode tracking, PM_Inv employs the **Predominant Mode (PM)** concept, in which the theoretical dispersion branch exhibiting the maximum surface-wave amplitude at each frequency is automatically selected during the inversion process. This strategy eliminates subjective modal indexing and enables a fully automated inversion workflow for complex multimodal dispersion datasets.

The software implements the **Predominant Mode (PM) inversion framework**, supporting both:

- **Vertical Component Predominant Mode (VCPM)** inversion
- **Radial Component Predominant Mode (RCPM)** inversion

The inversion can be performed using either:

- **PM_Inv_Input_GUI.m** *(recommended)* – graphical user interface
- **PM_Inv_Input.m** – script-based interface

Both interfaces execute the same inversion engine and produce identical results.

This software accompanies the publication:

[1]. Bhaumik, M., & Cox, B. R. (2026). Radial-Component Predominant-Mode Inversion of Rayleigh Waves: Application to DAS-based Site Characterization. *arXiv* preprint arXiv:2605.16717. https://doi.org/10.48550/arXiv.2605.16717

[2]. Bhaumik, M., & Cox, B. R. (2026). Predominant-mode inversion of surface waves: Inherently addressing inconspicuous low frequency mode jumps. *Engineering Geology*, 108834. https://doi.org/10.1016/j.enggeo.2026.108834

[3]. Bhaumik, M., & Naskar, T. (2024). Computation of Surface Wave's Dominating Mode for Stratified Media. *Indian Geotechnical Journal*, *55*(3), 1699-1714. https://doi.org/10.1007/s40098-024-01045-x

2. Requirements and Installation

Software Requirements

PM_Inv requires:

- MATLAB **R2019a** or later
- **Parallel Computing Toolbox** (recommended)
- **Statistics and Machine Learning Toolbox**

The inversion engine utilizes parfor loops when the Parallel Computing Toolbox is available. If the toolbox is not installed, the software automatically executes in serial mode (slow).

No compilation is needed. Keep the folder structure intact - the GUI adds the helpers subfolder to the MATLAB path at start-up and checks that the required toolboxes are installed, pointing you to the installers if any is missing.

3. Quick Start (GUI)

The graphical user interface is recommended for most users.

1. Open MATLAB.
2. Navigate to the PM_Inv directory.
3. Run : PM_Inv_Input_GUI
4. Complete the input fields sequentially from the leftmost tab to the rightmost tab.
5. Hover the mouse pointer over any input field to display its description.
6. Click **Preview Target** to verify that the dispersion curve has been imported correctly.
7. Click **Run Inversion** to start the optimization.

During the inversion, the software displays real-time updates of:

- inversion progress
- current best model
- misfit evolution
- predicted dispersion curve

All output files are automatically saved in the specified output directory upon completion.

4. The GUI, Tab by Tab

4.1 Target Data

Defines the observed data to be fitted.

- **Dispersion file (fv):** path to a .txt/.csv/.xlsx file. Column 1 = frequency (Hz); column 2 = phase velocity (m/s) or slowness; column 3 = optional uncertainty (see below).
- **3rd-column meaning:** tells the code what column 3 is: COV (coefficient of variation) or STD (standard deviation). None- the column is used as the velocity standard deviation as-is; if the file actually holds a COV, leaving this at (none) makes the 1/std^2 weights explode. Click **Preview Target** to verify that the dispersion curve has been imported correctly.

- **Measured_component:** Specifies the surface-wave component used for inversion. Available options:
  - VCPM – Vertical Component Predominant Mode
  - RCPM – Radial Component Predominant Mode
- **Include_min_COV:** used only when the file has no 3rd column; builds a std column as velocity times this COV (e.g. 0.05).
- **Resample / Number of log samples:** optionally resample the target onto a logarithmic frequency grid (set yes and the number of points).

4.2 Layering and Model

Controls the model parameterization.

- **Thickness mode (h):** variable (layer thicknesses inverted) or fixed (constant thicknesses).
- **h_range:** LN_based - h_count is the number of layers; LR_based - h_count is the thickness ratio between consecutive layers (Cox and Teague, 2016).
- **Layering (if fixed):** increasing or equal layer thicknesses (used only when h = fixed).
- **h_count:** batch input - a single value or several (e.g. 5 6 7, 5:7, or 5,6,8). Each value runs as a separate case and produces its own results.
- **Vs / nu / rho:** each has a mode (variable/fixed) and a range. For variable give two values min max; for fixed give one value. Vs_range also accepts Auto (bounds estimated from the data).

You will be able to manually change or constrain the limits later in the 'Layer Limits' tab.

4.3 Constraints and Optimizer

- **Vs_reversal:** maximum allowed fractional Vs drop between consecutive layers (velocity-inversion control). This parameter controls the occurrence of velocity reversals in the inverted profile. Use low values (≤ 0.1) for sites expected to have a monotonically increasing Vs profile to suppress spurious velocity reversal. Use larger values (0.5–1.0) only when the measured dispersion curve provides clear evidence of a low-velocity layer or velocity reversal.
- **Vs_rev_lim:** depth below which velocity reversals are not allowed. Velocity reversals are permitted only above this depth, while all deeper layers are constrained to exhibit non-decreasing shear-wave velocity.
- **Vs_halfspace_max:** force the half-space to be the fastest layer (yes/no).
- **DCR_range:** depth-conversion ratio setting the maximum investigation depth, d_max = max(wavelength)/DCR. Typical values range from 2 to 3, corresponding to a maximum investigation depth of approximately one-half to one-third of the maximum measured wavelength, respectively.
  - A **single value** specifies a fixed maximum investigation depth.
  - **Two values** define a range from which the DCR is randomly selected during model initialization, increasing the diversity of the initial population.
- **DE controls:**

  These parameters control the optimization algorithm.

- **Trial_num** – Number of independent inversion trials performed for each case. A single integer (e.g., 1, 2, or 3) runs the corresponding trial, while index range (e.g., 1:3) executes multiple trials sequentially in batch mode.
- **MaxItr** – Maximum number of generations (iterations) in each trial. Larger values generally improve the convergence but increase the computational time. Values between 50 and 100 are recommended for most applications.
- **NP** – Population size. A larger population provides greater model diversity and may improve exploration of the search space, but increases the computational cost. Population sizes between 50 and 100 are generally recommended.
- **CR** – Crossover probability. Recommended: 0.8–1.0.
- **pbestFrac** – Fraction of the best-performing individuals used in the JADE mutation strategy. Recommended: 0.10–0.20.
- **TolStall** – Convergence tolerance used for early stopping. The optimization terminates when the improvement in the objective function remains below this threshold for a specified number of generations. Recommended: 1×10^{-7} to 1×10^{-5}.
- **StallGens** – Number of consecutive generations satisfying the convergence criterion before terminating the optimization. Recommended: 10–30.

- **HTLM controls:** d (polynomial order), EPW (elements per wavelength), PML (absorbing-boundary layers) - trade forward-model accuracy against speed. These parameters are recommended to remain at their default values unless modifications are required for advanced numerical studies or method development.

4.4 Layer Limits (optional)

This tab allows the user to manually refine the search bounds for individual layers. Click Compute to automatically generate the lower and upper bounds for each layer based on the selected inversion settings. The generated bounds are displayed in the editable table. Users may modify any lower or upper bound to further constrain or relax the search space for a particular layer. The edited limits are used during both Preview and Run Inversion.

If no changes are made, the software uses the automatically computed layer limits.

4.5 Figures and Control

This tab displays real-time inversion results and provides controls for monitoring and managing the optimization process. During the inversion, the following panels are updated continuously:

- Measured and predicted dispersion curves
- Inverted Vs-depth profile (full-depth and zoomed views)
- Differential Evolution convergence history

The **Pause** button temporarily suspends the optimization and allows it to be resumed later, while **Stop** terminates the inversion prematurely. In both cases, all results generated up to that point are automatically saved to the output directory.

4.6 Post-Processing

This tab provides visualization and analysis of previously saved inversion results without rerunning the inversion.

- **Select .mat:** pick one or more saved result files.
- **Mode:** Best (single lowest-misfit model per h_count), All (the Nbest lowest-misfit models from each trial), or MeanBest (mean of the best-per-trial models per h_count).
- **Dmax / Nbest / Modes:** plotting depth, models per trial (All mode), and number of theoretical modes overlaid.
- **Plot:** draws two panels - dispersion (modes and predominant mode versus the target) and Vs profiles.

4.7 Action Buttons

- **Preview Target:** load, preprocess and plot the target only.
- **Export Config:** push Target/DE/HTLM/Layer_param structs to the base workspace for scripting.
- **Run Inversion:** run the full inversion inline.

5. Outputs

Results are written to a new subfolder named after the target file. It contains:

- **input_params.mat:** the Target, Layer_param, HTLM and DE settings used.
- **Temp_params.mat:** the initial model and parameter bounds for the current case.
- **name_h_count_H_trial_T.mat:** one file per case, holding the best model (xbest), the full per-generation history (param), the number of layers, the bounds, and the target.

The status bar and Command Window print "Running VCPM Profile: ..." (or RCPM) with the current h_count and trial.

6. Script Usage (alternative)

Open PM_Inv_Input.m, edit the parameters in the clearly marked blocks at the top (Target, DE, HTLM, Layer_param), and press Run. The script performs the identical inversion and saves the same outputs. PM_Inv_PostProcessing.m plots them - select the .mat files when prompted.

7. Tips and Troubleshooting

- **Weights blow up / huge misfit:** Ensure the third column of the dispersion file is interpreted correctly. If it contains COV, set Third-column Interpretation = COV. Using an incorrect interpretation may result in unrealistic weighting during the inversion.
- **Results look unstable:** The inversion is sensitive to the chosen model parameterization. Compare results obtained using multiple **h_count** values and several independent inversion trials. Use the **Best** and **All** options in the **post-processing** tab to assess the consistency of the inverted profiles.
- **Slow runs:** For improved performance, use a workstation equipped with the **Parallel Computing Toolbox**. Computational time can be reduced by decreasing **HTLM** parameters (**d** and **EPW**) or the Differential Evolution parameters (**NP** and **MaxItr**), although this may reduce forward-model accuracy or optimization performance.

- **Depth range:** control the maximum inverted depth with DCR_range and the plotting depth with Dmax.
- **GUI survives close all:** the input window is protected; analysis figures may still be closed by the run loop.

8. Contact and Support

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Questions, feedback, bug reports, and collaboration proposals are welcome.

PM_Inv is an actively developed research software package and will continue to evolve with ongoing research. Future releases may include bug fixes, performance improvements, additional inversion capabilities, and new features. Users are encouraged to check the GitHub repository periodically for the latest updates and improvements.

This software is provided **"as is"**, without warranty of any kind, express or implied. While every effort has been made to verify the correctness of the implementation, the authors assume no responsibility for errors, omissions, or any consequences arising from the use of this software.

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